

Technology Trends Analysis Based on Stack Overflow Developer Survey

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OUTLINE



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EXECUTIVE SUMMARY



- This project analyzes technology trends using Stack Overflow Developer Survey data.
- The analysis identifies the top programming languages and databases currently in use.
- Results highlight strong demand for data-related and cloud-based technologies.
- Future trends indicate increasing interest in Python, cloud platforms, and scalable databases.
- These insights support informed decision-making for career planning, hiring, and technology investments.



INTRODUCTION



- Purpose: To analyze current and future technology trends using Stack Overflow Developer Survey data.
- Target Audience: Students, job seekers, educators, and technology decision-makers.
- Value: The analysis provides insights that support informed decisions related to career development, hiring, and technology adoption.



METHODOLOGY

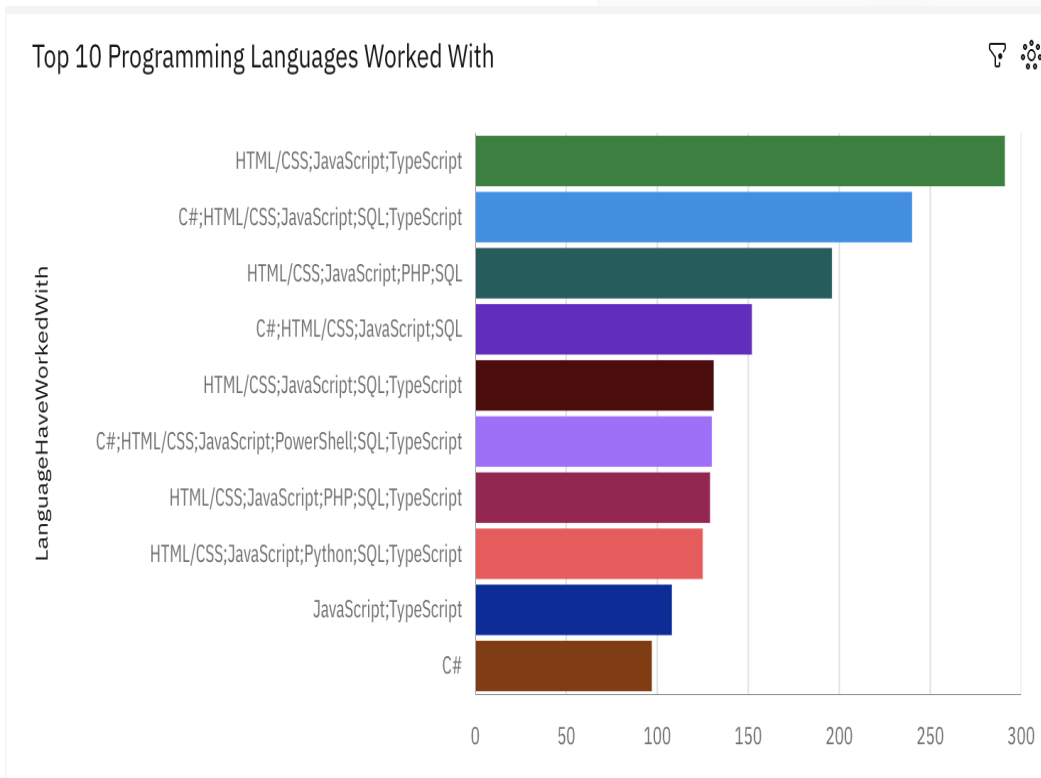


- Data Source: Stack Overflow Developer Survey dataset.
- Data Collection: Survey-based data collected from developers worldwide.
- Data Wrangling: Data cleaning included removing missing values, filtering relevant variables, and selecting top 10 technologies.
- Tools Used: IBM Cognos Analytics for data visualization and dashboard creation.



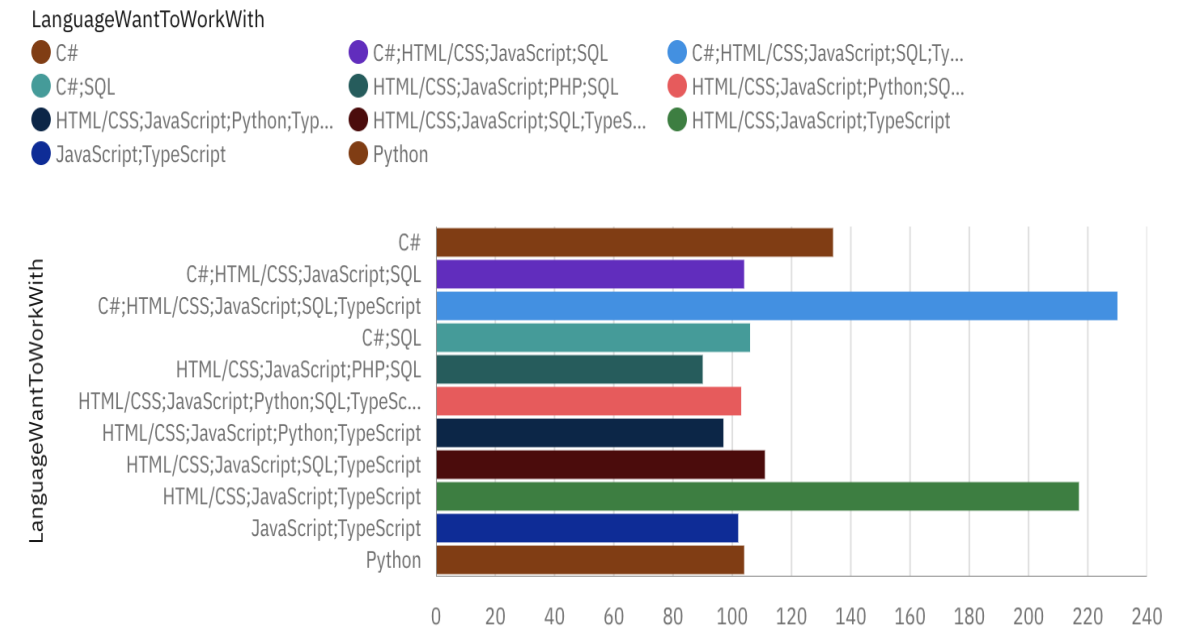
PROGRAMMING LANGUAGE TRENDS

Current Year



Next Year

Top 10 Programming Languages to Work With



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- Web development technologies such as HTML/CSS, JavaScript, and TypeScript dominate current programming language usage.
- Python shows strong and growing popularity, especially in future technology trends.
- Developers increasingly report using multiple programming languages together, indicating a multi-skill development environment.

Implications

- Learning JavaScript and TypeScript remains essential for front-end and full-stack development roles.
- The rising interest in Python highlights its importance for data analytics, automation, and AI-related roles.
- Developers and organizations should focus on versatile skill sets rather than a single programming language to stay competitive.

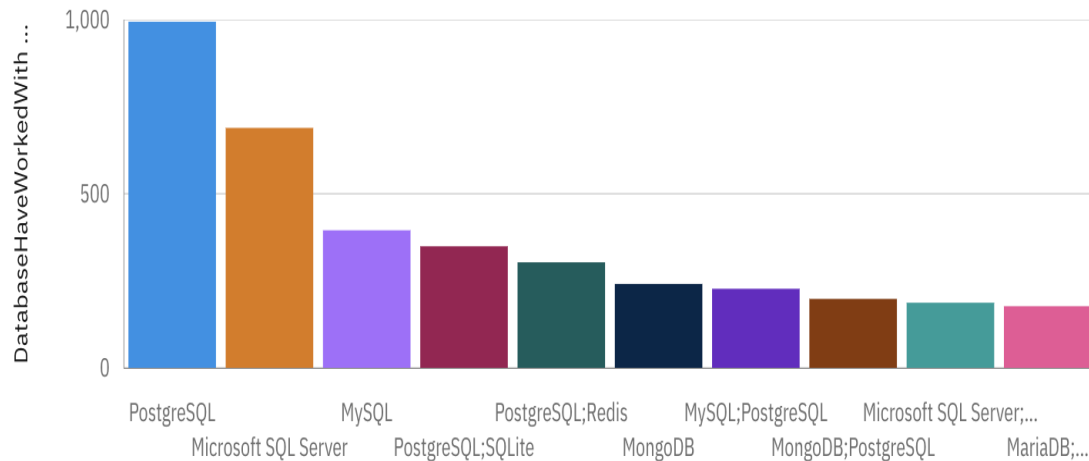
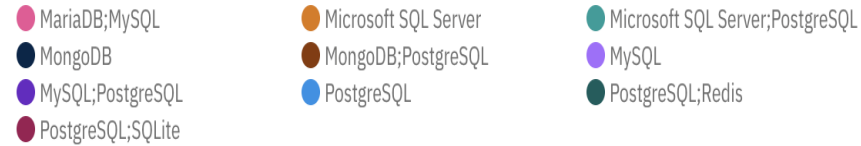


DATABASE TRENDS

Current Year

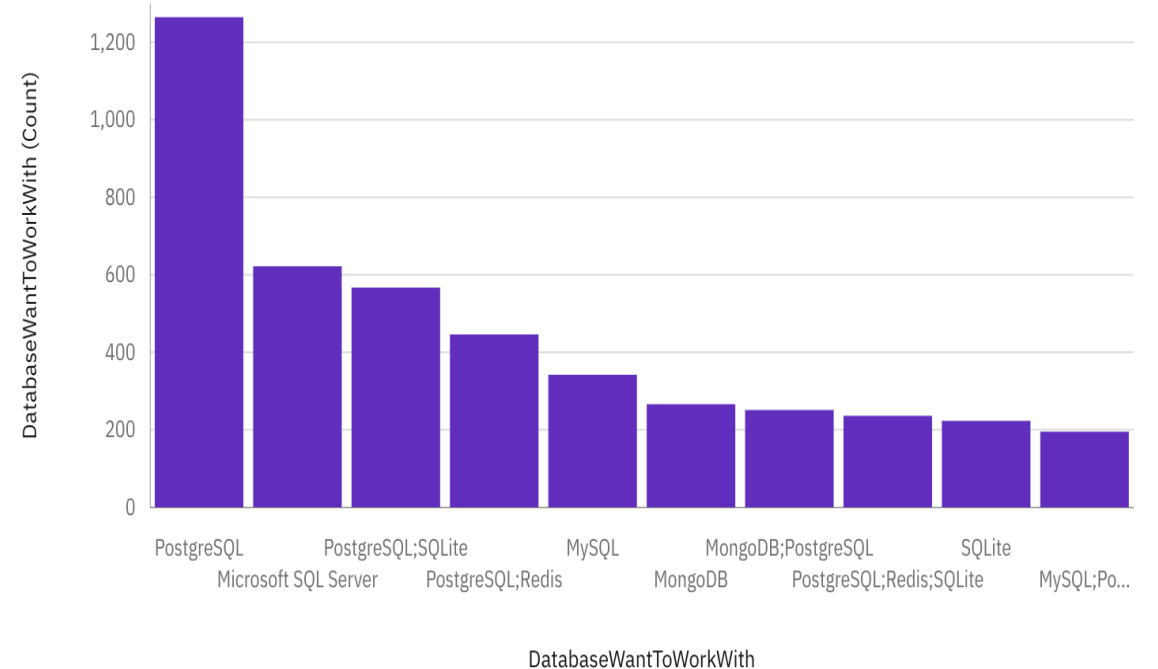
Top 10 Databases Worked With

DatabaseHaveWorkedWith



Next Year

Top 10 Databases to Work With



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- Relational databases such as **PostgreSQL** and **MySQL** are the most widely used databases in the current technology landscape.
- **PostgreSQL** shows strong growth in future demand, indicating increasing preference for scalable and open-source database solutions.
- **NoSQL** databases like **MongoDB** continue to gain interest for handling flexible and unstructured data.

Implications

- Strong demand for **PostgreSQL** and **MySQL** suggests that database fundamentals remain critical for most development roles.
- The rising popularity of **PostgreSQL** highlights the importance of learning advanced **SQL** and performance optimization skills.
- Developers should gain exposure to both relational and **NoSQL** databases to meet diverse industry requirements.



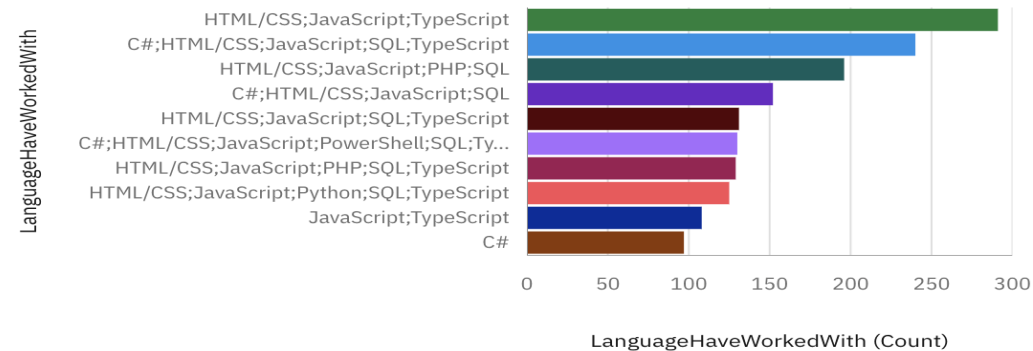
DASHBOARD



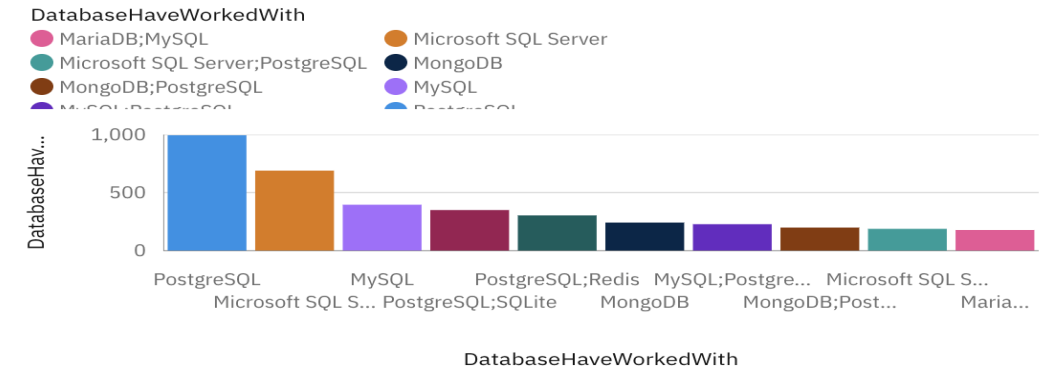
<Please present your dashboard in the following slides.>

DASHBOARD :Current Technology Usage

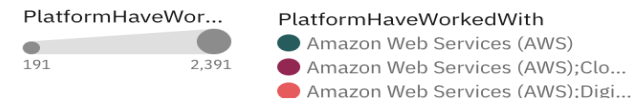
Top 10 Programming Languages Worked With



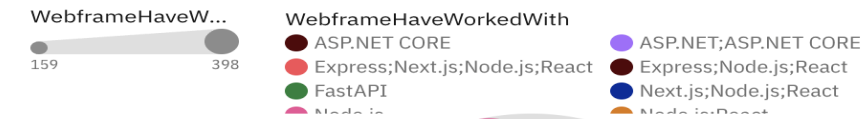
Top 10 Databases Worked With



Top 10 Platforms Worked With

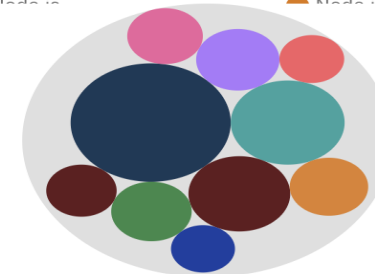


Top 10 Web Frameworks Worked With

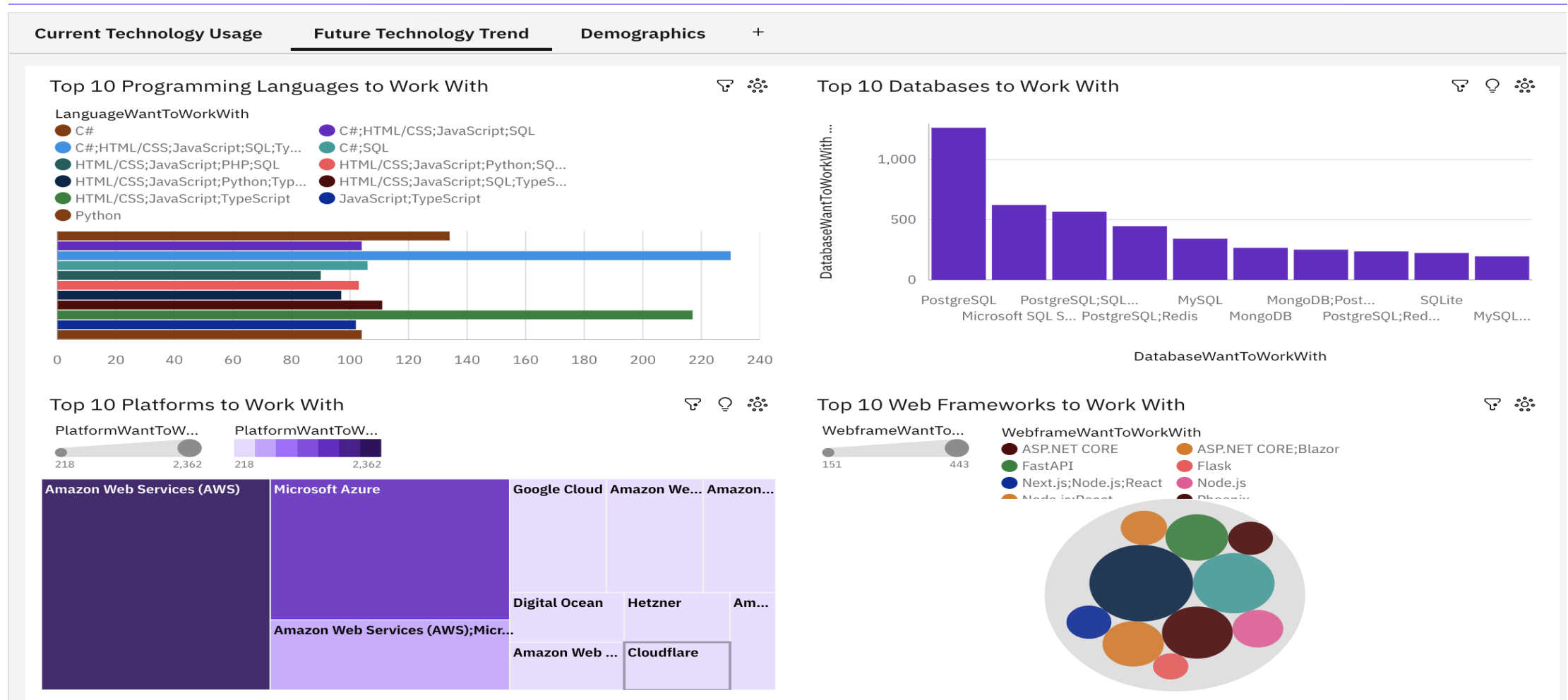


Amazon Web Services (AWS)

Microsoft Azure
Amazon Web Services (AWS);Microsoft Azure



DASHBOARD : Future Technology Trends



DASHBOARD: Demographics

Current Technology Usage

Future Technology Trend

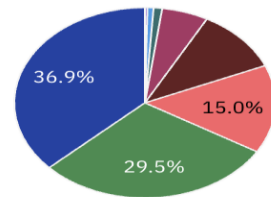
Demographics

+

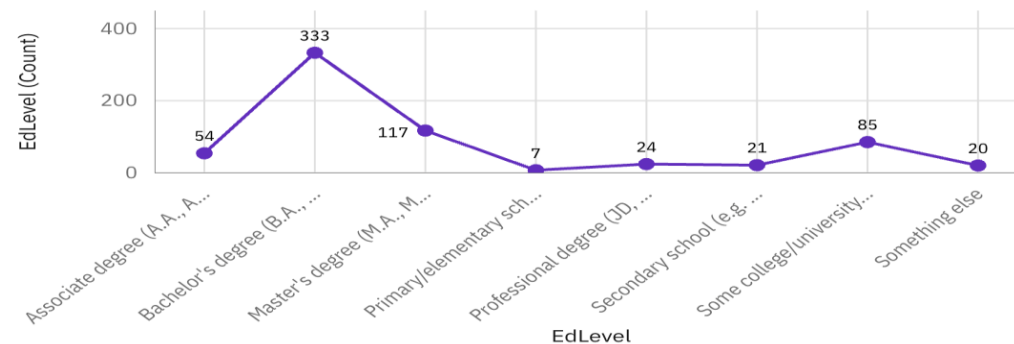
Respondent Distribution by Age

Age

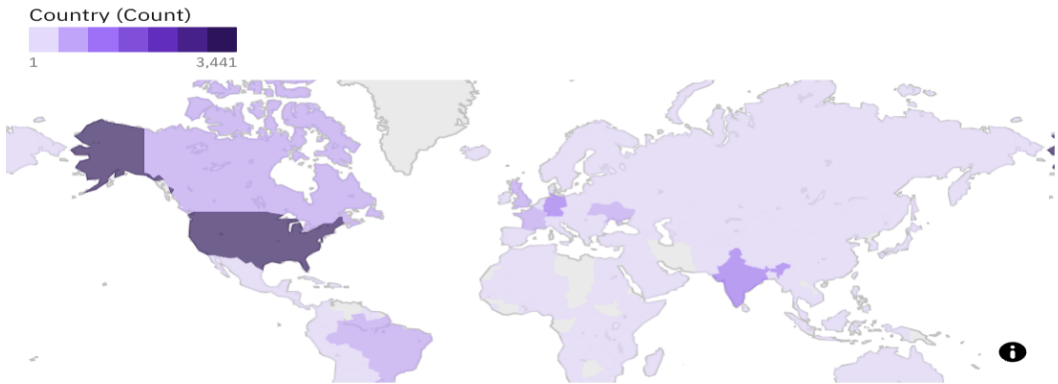
- Prefer not to say
- 18-24 years old
- 65 years or older
- 45-54 years old
- Under 18 years old
- 35-44 years old
- 55-64 years old
- 25-34 years old



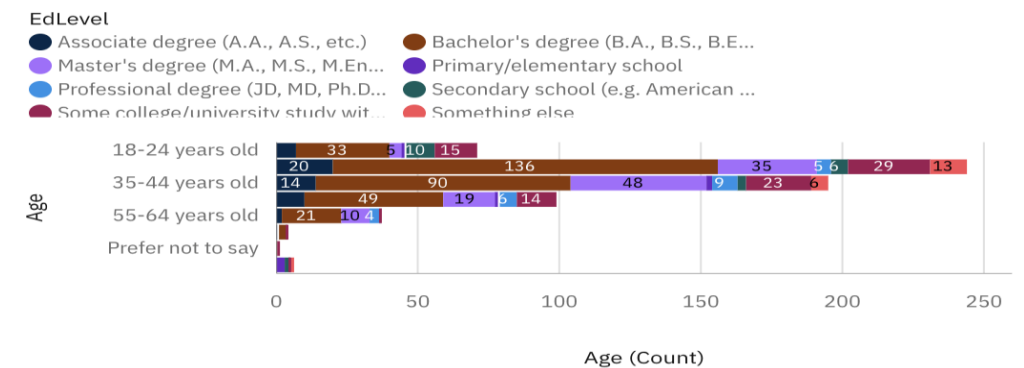
Respondent Distribution by Formal Education Level



Respondent Count by Country



Respondent Count by Age and Education Level



DISCUSSION



- Web technologies remain dominant, confirming continued demand for front-end and full-stack development skills.
- Python's growing popularity reflects its expanding role in data analytics, automation, and AI.
- Relational databases such as PostgreSQL and MySQL continue to lead, with increasing adoption of NoSQL for flexible data needs.
- Cloud platforms like AWS and Azure highlight the shift toward scalable, cloud-based solutions.



OVERALL FINDINGS & IMPLICATIONS

Findings

- Web development technologies remain the most widely used across the developer ecosystem.
- Python and PostgreSQL show strong growth in future demand, driven by data and AI use cases.
- Cloud platforms and multi-technology stacks are becoming standard in modern development.

Implications

- Developers should focus on full-stack skills combining web, databases, and cloud technologies.
- Learning Python and advanced SQL provides strong career advantages in analytics and automation.
- Organizations should prioritize scalable, cloud-ready architectures and versatile skill sets.



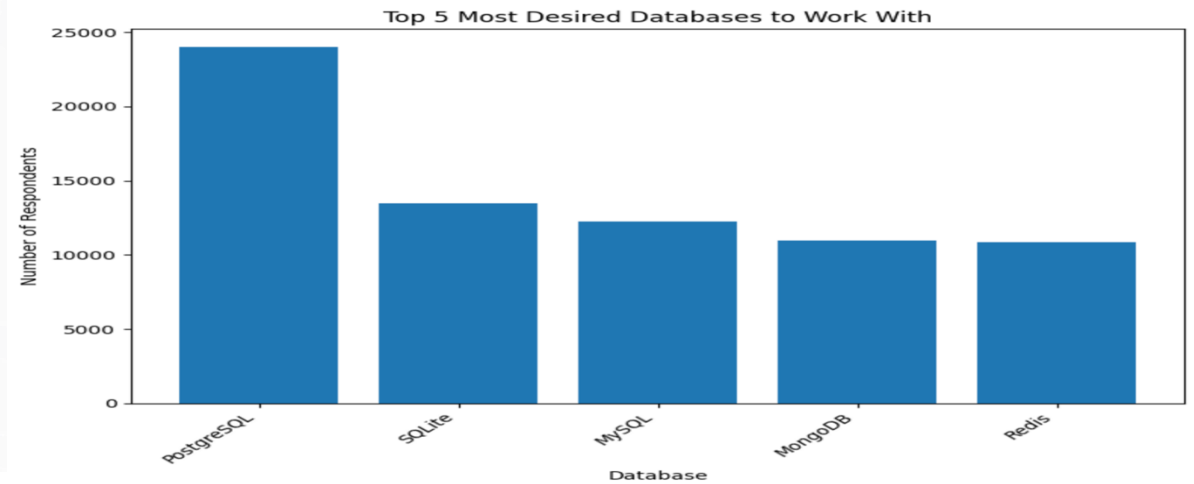
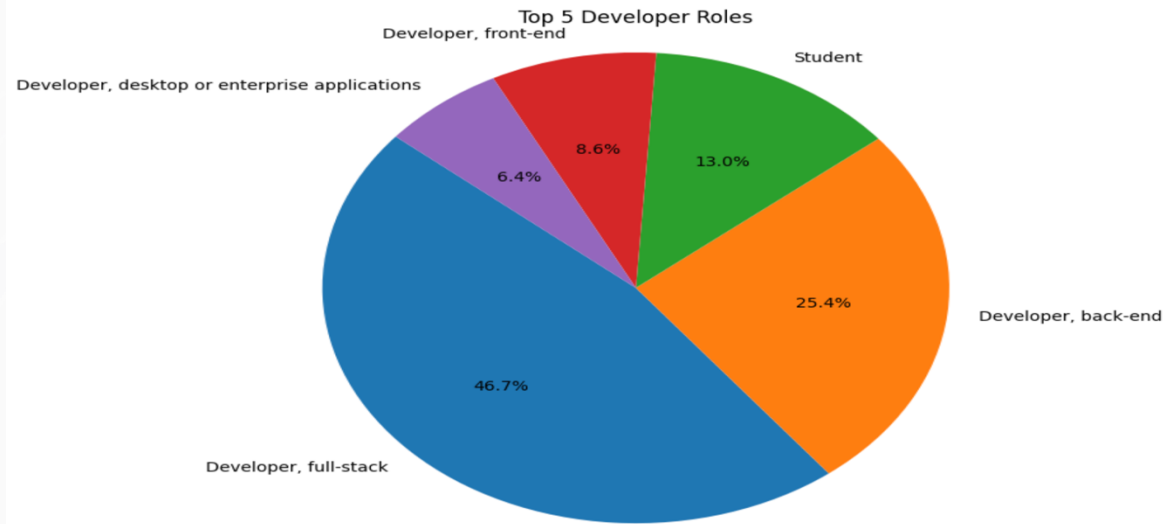
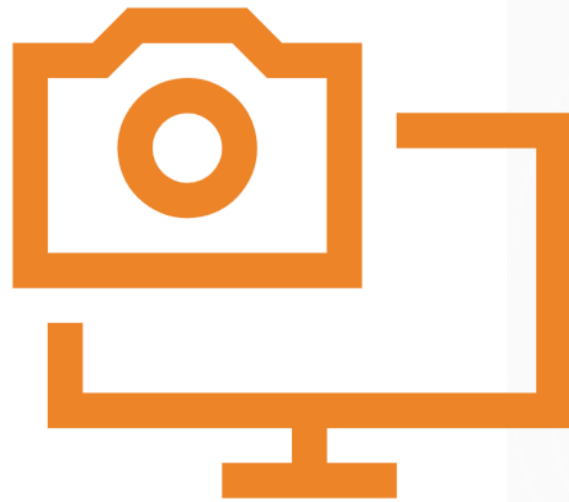
CONCLUSION



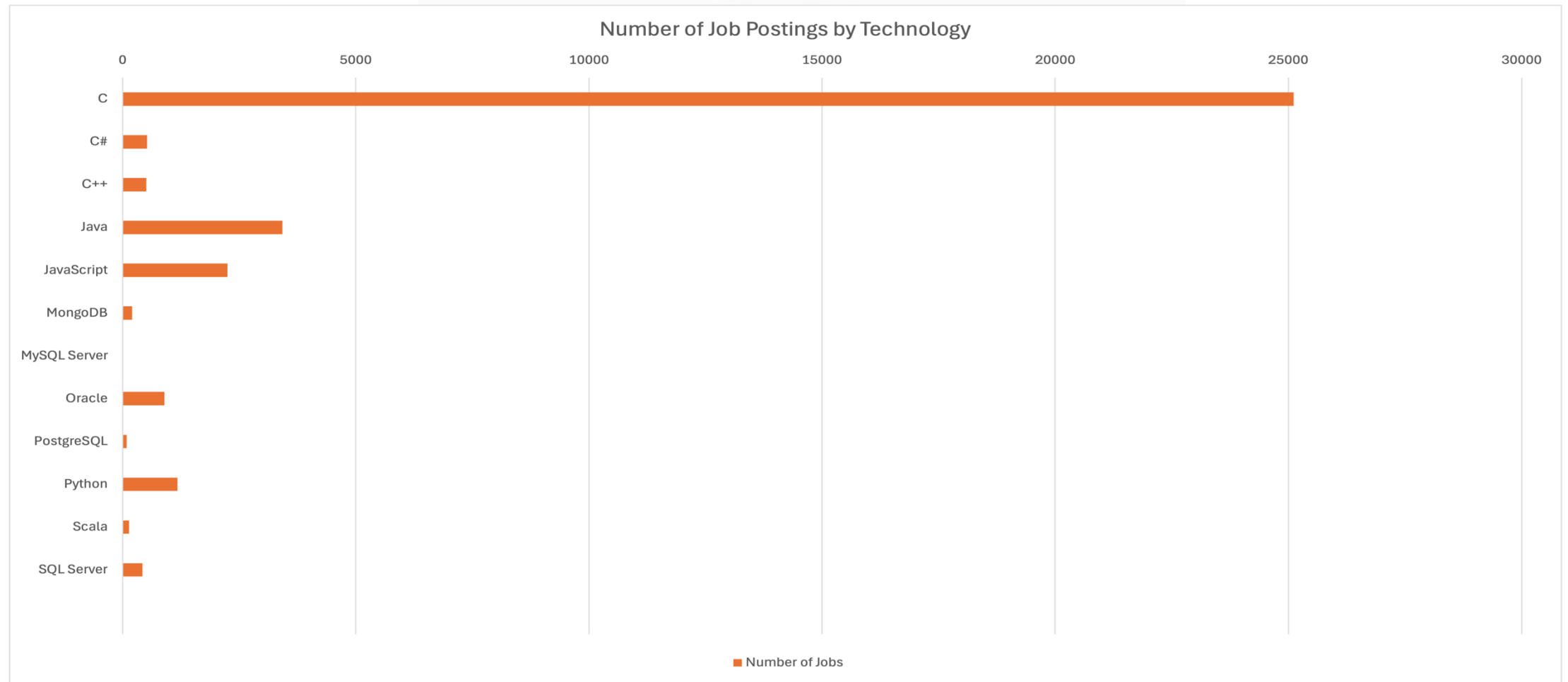
- The analysis shows that web technologies, databases, and cloud platforms continue to form the core of modern software development.
- Python and PostgreSQL stand out as high-growth technologies, driven by data analytics, automation, and AI use cases.
- Developers and organizations that invest in versatile, full-stack and cloud-ready skill sets will be better positioned for future technology demands.



APPENDIX



JOB POSTINGS



POPULAR LANGUAGES

